

1 Overview

Software engineers make important decisions every day, such as the priority and severity of security vulnerabilities [1] or the estimated cost of implementing a user story or fixing a defect [2]. Such decisions are crucial to software maintenance and agile sprint planning [3], but are often difficult and time-consuming to make [2,4]. Machine learning techniques have been applied to model such software engineering (SE) decisions to help reduce human effort. However, there are two main challenges in modeling these decisions. **Challenge 1 Efficiency:** these SE decisions are subjective and contextual – a model trained on the decisions of one project does not accurately predict the decisions of other projects [2,5]. Therefore, project-specific models must be trained on a sufficient amount of training data before they can be applied to predict SE decisions on the same project. Reducing the human effort required to generate such training data becomes the key to the success of modeling SE decisions. **Challenge 2 Biases:** algorithmic bias of the machine learning model is another threat to the success of modeling SE decisions. For example, overestimating the cost of certain types of tasks or underestimating the severity of certain types of vulnerability can have severe and unfair consequences.

In order to solve the above two challenges, this work explores how to accurately model project-specific SE decisions with increased efficiency (less human effort required to generate a sufficient amount of training data) and reduced bias (model predictions should not be more accurate in or favoring one subgroup of data over another). To simplify our discussion, we will use the story point estimation problem as a case study in the rest of this proposal to demonstrate this work.

Story point estimation is an essential part of agile software development. Story points are unitless, project-specific effort estimates that help developers plan their sprints [6,7]. Traditionally, developers estimate story points collaboratively using planning poker or other manual techniques [8]. Although initial calibration of the estimates for each project is helpful, once a team has converged on a set of precedents, story point estimation can become tedious and labor intensive [9]. State-of-the-art approaches [9,10] tackle this problem by training machine learning models to predict story points. However, because the story points estimated by humans are highly subjective and contextual, such machine learning models perform poorly in cross-project predictions [9,10] – these human decisions from individual projects are not transferrable to other or new projects. As a result, prediction only works within a project and a sufficient amount of human decisions are still required as training data for each new software project.

1.1 Intellectual Merit

We will explain the three research thrusts explored in this work using the case study of story point estimation mentioned above. Figure 1 shows an overview of the process.

Thrust 1: Efficiency. To increase efficiency and reduce human effort, we propose to model developers’ estimates from comparative judgments instead of direct story points. That is, instead of asking the developers what the story point of item A is, we will ask whether item A or B requires more effort to implement. According to the law of comparative judgment [11] in Psychology, providing such comparative judgments yields a lower cognitive burden for humans than providing ratings or categorical labels. To thoroughly test the feasibility and benefits of learning from comparative judgments, we plan to explore the three research questions in this research thrust:

- **RQ1.1 Can models trained on comparative judgments predict as accurately as models trained on direct ratings?** Given the presumable cost-saving advantage of comparative judgments, we will develop a novel learning framework to train machine learning models on comparative judgments and evaluate whether they can achieve similar performance as the regression models. Preliminary results on story point estimation (shown later in Section 2.2) suggest that

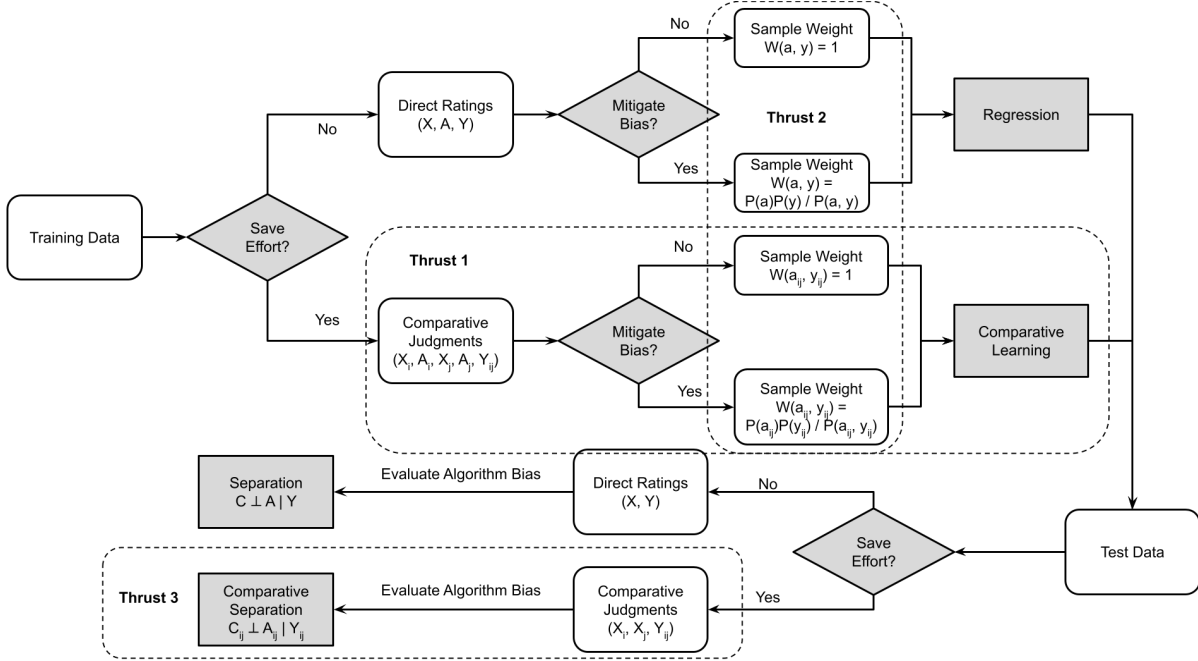


Figure 1: Efficient and unbiased modeling of software engineering decisions.

the answer is yes. This serves as the basis for the proposal.

- **RQ1.2 Does comparative learning require less human annotation effort than regression models to achieve similar performance?** Although the law of comparative judgment [11] has indicated that the cost of comparative judgments is lower than that of direct ratings, we need real-world cost data to analyze the cost-saving benefits of the proposed approach. To this end, we plan to collect story point estimation data from students reporting the time spent providing comparative judgments or direct story point estimates. This data will be the first to compare the cost of comparative judgments and direct estimates of story points.
- **RQ1.3 Does the model benefit from actively choosing which comparative judgments to query?** Active learning [12] has been widely applied in modeling direct human ratings and has shown a great advantage in reducing human effort. In this research, we will be the first to explore how much cost can be saved by actively selecting the most informative comparative judgment to query next with strategies such as uncertainty sampling.

Thrust 2: Bias reduction. Biases in SE decisions, such as underestimating the difficulty of implementing a specific type of user story (by assigning lower story points), can have severe and unfair consequences, such as overloading developers working on those user stories or overlooking their work. Therefore, it is important to reduce possible algorithmic biases when modeling these decisions. Although fairness and bias issues have been extensively explored for binary classification problems, there are few studies on how to reduce algorithmic bias for regression models (such as a story point estimation model). Among the different definitions of algorithmic bias, we focus specifically on satisfying separation [13] – the model’s predictions are conditionally independent from the sensitive attribute given the dependent variable $C \perp A | Y$:

- **RQ2.1 How to ensure better separation for regression models?** We propose a novel preprocessing technique FairReweighting inspired by the Reweighting algorithm [14] in binary classification to ensure the learned regression model satisfies separation. Both theoretically and empirically, we will show that the proposed approach reduces bias of the regression model.

- **RQ2.2 How to ensure better separation for comparative models?** We will explore how to adapt the proposed FairReweighting algorithm in RQ2.1 to comparative judgments. The learned model will be evaluated on a test set with ground truth story points for its algorithmic bias in terms of separation. This would be the first bias mitigation algorithm for comparative learning.

Thrust 3: Bias evaluation in comparative judgments. In addition to saving effort by using comparative judgments as training data, human effort can be further reduced by constructing test data with comparative judgments. However, the current bias evaluation metrics for separation are all based on classification or regression labels. Therefore, in this research thrust, we propose a novel notion of comparative separation which is evaluated with comparative judgment labels. We will analyze how this comparative separation is related to separation in the following research questions:

- **RQ3.1 Is comparative separation equivalent to separation in binary classifications?** We will show that comparative separation is equivalent to separation in binary classifications.
- **RQ3.2 In binary classification, how much effort can be saved by using comparative separation instead of separation to evaluate algorithmic bias?** We will analyze the relationships between the variances of the bias metrics and the size of the test set. That is, how many comparative judgments or direct classifications are required to achieve the same level of evaluation variance. Together with the result of **RQ2.1**, we can compare the effort required to test comparative separation vs. test separation in binary classifications.
- **RQ3.3 What is the relationship between comparative separation and separation in regression?** In this research question, we will explore whether comparative separation is still equivalent to separation when the dependent variable is numerical.

1.2 PI Qualification

For Thrust 1, the PI has extensive experience in developing active learning models to assist human decision making by (1) reducing human effort, (2) estimating the required human effort before hand, and (3) detecting potentially incorrect human decisions. The PI has explored many scenarios in software engineering and has achieved state-of-the-art performance in citation screening [15], self-admitted technical debt detection [17], software security vulnerability inspection [18], and actionable static warning identification [19,20]. For Thrust 2, the PI has a related work showing that equalized odds (separation in binary classification) can be achieved when the model is trained with balanced classes in each sensitive group [21]. The PI has also received an NSF CRII award for detecting unfairness in human decisions with machine learning software. Preliminary results from the CRII award (shown later in Section 2.2) showed the feasibility of learning from comparative judgments. These laid the foundation of both research thrust 1 and 2.

2 Research Plan

The research plan will be presented with respect to the story point estimation case study for easier demonstration. Note that the actual work will include experiments on other datasets as well.

2.1 Case Study: Story Point Estimation

2.1.1 Public Data

We will use the story point estimation dataset introduced in Choetkiertikul et al. [10] for experiments. The dataset consists of the titles and descriptions of the backlog items, as well as their assigned story points collected through JIRA in 16 projects. During the pre-processing step, the titles and descriptions are concatenated as the final text feature [9]. Shown in Figure 2 (a), we specifically used the Jira Software project for preliminary experiments by separating the project items into two sensitive groups: (1) the $A = 0$ group contains the user stories to implement and (2) the $A = 1$ group contains the defect fixing requests. Figure 2 (b) shows the comparative judgment data. In reality, Y_{ij} will be acquired by asking developers to compare the effort of implementing a

X	A	Y	Xi	Ai	Xj	Aj	Yij
As a user I would like the ability to see a horizontal swimlane on my Rapid Board	0	13	As a user I would like the ability to see a horizontal swimlane on my Rapid Board	0	Generic webwork aliases may clash with other plugins	1	1
Generic webwork aliases may clash with other plugins	1	5	Generic webwork aliases may clash with other plugins	1	URL doesn't select issue on the Planning Board as it should	1	1
As a user I would like the Rapid Board to be available via Labs	0	5	As a user I would like the Rapid Board to be available via Labs	0	As a user I would like the ability to see a horizontal swimlane on my Rapid Board	0	-1
URL doesn't select issue on the Planning Board as it should	1	3	URL doesn't select issue on the Planning Board as it should	1	As a user I would like the Rapid Board to be available via Labs	0	-1

(a) Direct ratings.

(b) Comparative judgments.

Figure 2: Story point estimation with direct ratings and comparative judgments.

given pair of items (X_i, X_j) :

$$Y_{ij} = \begin{cases} 1, & \text{when } x_i \text{ requires more effort than } x_j \\ -1, & \text{when } x_i \text{ requires less effort than } x_j. \end{cases}$$

Here, we simulate this process with $Y_{ij} = \text{sgn}(Y_i - Y_j)$.

2.1.2 Problem Statements

- **Thrust 1:** Given the comparative judgments (X_i, X_j, Y_{ij}) as training data, learn a model $C = f_\theta(x)$ to accurately predict the story point Y of each item $x \in X$.
- **Thrust 2:** Given the binary sensitive attribute $A \in \{0, 1\}$ indicating the type of backlog items, ensure that the learned model $C = f_\theta(x)$ is not biased towards a specific type $A = a$ by the standard of separation $C \perp A|Y$.
- **Thrust 3:** Given a test set of comparative judgments (X_i, X_j, Y_{ij}) , evaluate whether model $C = f_\theta(x)$ satisfies separation.

2.1.3 Generalizability

Note that the proposed work is not restricted to this story point estimation problem but can be applied to various scenarios. This case study is chosen because of its availability of data, as well as its similarity with many other decision-making processes such as software vulnerability severity assessment, face beauty rating, talent hiring, school admission, etc. Furthermore, biases in story point estimation can be analyzed without raising much ethical concern.

2.2 Thrust 1: Efficiency

The goal of Thrust 1 is to reduce the effort of human annotation to learn an accurate model $C = f_\theta(x)$ to predict the estimated story points Y . We plan to approach this thrust by learning from comparative judgments and with active learning.

2.2.1 Background

Active Learning is a human-in-the-loop learning strategy. It actively selects which labels to query based on the current model's need [12]. Previous work of the PI has applied different query strategies from active learning to increase efficiency (up to 90% human effort saved) in various software engineering tasks including searching for better software configurations [22], inspecting for software vulnerabilities [23], identifying self-admitted technical debts [17], prioritizing test cases [24], semi-automating data annotations [25], recognizing actionable static code warnings [19, 20], screening for relevant literature [26]. For the story point estimation problem, the same active learning-based solution with pre-trained transformer models such as Sentence-BERT [27] would be a good baseline solution to reduce human effort.

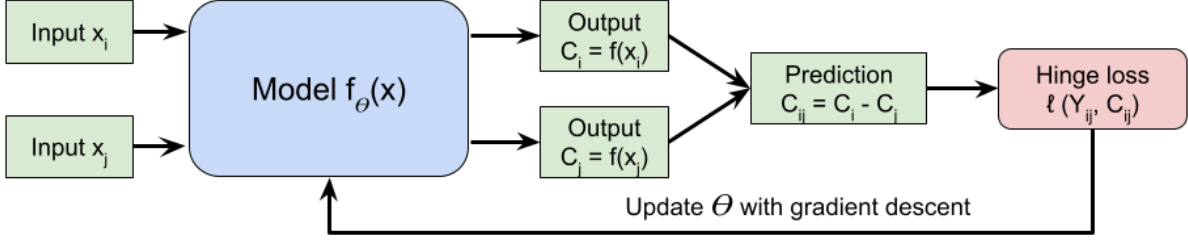


Figure 3: The comparative learning framework.

Comparative judgments query for preferences on a pair of data points instead of directly rating scores (or binary decisions) on one data point. According to the law of comparative judgment [11] in Psychology, providing such comparative judgments yields a lower cognitive burden for humans than providing ratings or categorical labels. This could potentially reduce human effort due to lower cognitive burden of each human decision made. However, research on model learning from comparative judgments is very limited. Until now, only the work of Qian et al. [28] has learned a model based on linear support vector machine (SVM) on comparative judgments. The linearity of the model restricted its usage in accurately modeling complex data.

Active Learning in Comparative judgments could further reduce human effort by actively selecting which data pairs are more useful when their comparative judgments are provided. Qian et al. [28] also explored a specific query strategy called orthogonal query which utilizes the linearity of the SVM model to find the next data pair orthogonal to the current model’s decision vector. Unfortunately, this orthogonal query strategy would not be feasible when the model is nonlinear.

2.2.2 Proposed work

RQ1.1: Can models trained on comparative judgments predict as accurately as models trained on direct ratings?

Shown in Figure 3, we propose a novel comparative learning framework to learn from comparative judgments Y_{ij} of input pairs (x_i, x_j) . In this framework, although not explicitly available, we assume that each data point is associated with a hidden utility score. The model $f_\theta(x)$ learns to map the input x to its hidden utility score through gradient descent with hinge loss:

$$\ell(Y_{ij}, C_{ij}) = \max(0, 1 - (Y_{ij} \cdot C_{ij})) \quad (1)$$

Given a certain comparative judgment oracle $Y_{ij} = 1$ we only know that x_i requires more effort than x_j – the hidden utility score of x_i is higher than that of x_j . We do not know how much higher the utility score of x_i is. Therefore, hinge loss is appropriate since it only penalizes $C_{ij} = f_\theta(x_i) - f_\theta(x_j) < 1$. As long as $C_{ij} = f_\theta(x_i) - f_\theta(x_j) \geq 1$, we have $\ell(Y_{ij}, C_{ij}) = 0$.

Following this framework, $f_\theta(x)$ can be any regressor of any structure. This provides the proposed approach with more flexibility and generalizability than the linear SVM model in Qian et al. [28]. For the case study of story point estimation, the encoder $f_\theta(x)$ is chosen as a Sentence-BERT [27] pretrained on one billion sentence pairs with output layers replaced as a linear dense layer to predict the hidden utility score of each input x .

Evaluation Plan: Figure 4 shows the preliminary results on the Jira Software project [10]. Here, we compared the performance of a baseline regression model $f_\theta(x)$ of fine-tuned Sentence-BERT [27] (the blue solid line) with two models learned from comparative judgment data: (1) our proposed comparative learning approach fine-tuning the same Sentence-BERT model (the red dashed curve) and (2) a baseline LinearSVM-Comparative [28] model (the green dashed dot curve). Comparative

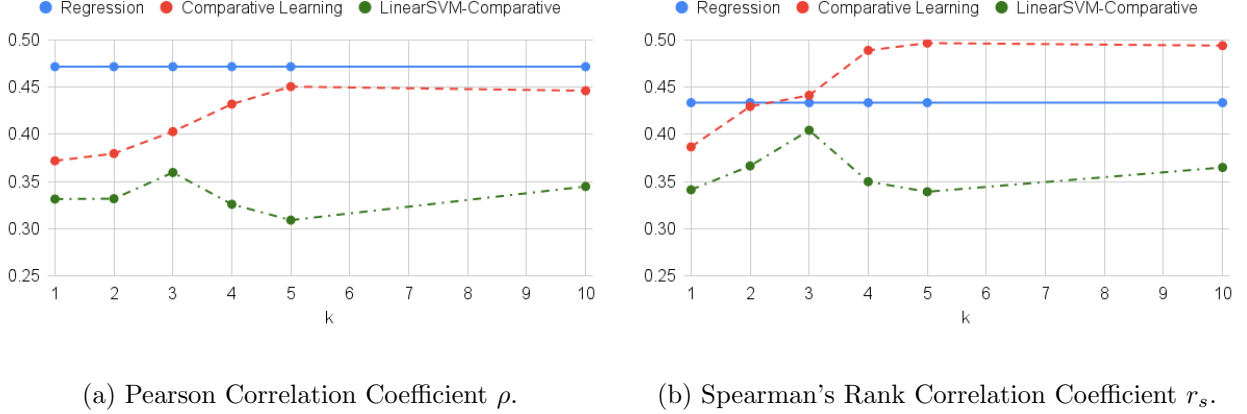


Figure 4: Preliminary results of RQ1.1 on Jira Software.

judgment oracles are constructed by randomly selecting k pairs for each item. Two metrics were applied to evaluate prediction performance. Pearson Correlation Coefficient [29] $\rho = \frac{\text{cov}(f_\theta(X), Y)}{\sigma_{f_\theta(X)} \sigma_Y}$ evaluates how closely the predicted utility function $f_\theta(x_i)$ aligns with the actual ratings y_i . Spearman's Rank Correlation Coefficient [30] $r_s = \rho [R[f_\theta(X)], R[Y]]$ evaluates how closely the rank of the predicted utility function $f_\theta(x_i)$ aligns with that of the actual ratings y_i .

Successful Criteria: RQ1.1 can be answered as yes if the proposed approach always outperforms LinearSVM-Comparative [28] and performs similarly to the regression model on most software projects. From preliminary results in Figure 4 on the Jira Software project we can observe:

- Our comparative learning approach always outperforms LinearSVM-Comparative [28].
- Our comparative learning approach achieves ρ as the baseline regression model. Especially when using more training data pairs (a larger k).
- Our comparative learning approach can even outperform the baseline regression model in terms of r_s with larger k . This is as expected since comparative judgments only provide order information.

This preliminary results suggest that it is very promising to learn from comparative judgments with the proposed framework. We plan to repeat this experiment on other projects of story point estimation as well as other case studies such as vulnerability severity prediction.

RQ1.2: Does comparative learning require less human annotation effort than regression models to achieve similar performance?

In order to compare the human effort required for learning from direct ratings and from comparative judgments, we plan to conduct human subject experiments on story point estimation. Forty students in the SE major will be equally divided into two groups.

Group 1: Each student will estimate around 20 story points Y_i directly. The time spent during this process will be recorded to calculate the average annotation time for a direct rating t_r .

Group 2: Each student will compare the effort of around 20 pairs of items to provide their comparative judgments Y_{ij} . The time spent during this process will be recorded to calculate the average annotation time for a comparative judgment t_c .

Evaluation Plan: For each rater's annotations, a comparative model or a regression model will be trained on the comparative oracles or the direct ratings respectively. Their performance will be tested in the same way described in RQ1.1.

Successful Criteria: RQ1.2 should be answered as yes when the performance of the comparative models is similar to that of the regression models, and their human effort required (measured by annotation time) is less— $t_c < t_r$.

Alternative Plan: If the performances of the comparative models are worse than those of the regression models and $t_c \times k < t_r$, we can train the comparative models with $k \times 20$ pairs – k pairs for each training data point. In this way, comparative models are expected to achieve better performance with more human annotation effort.

RQ1.3: Does the model benefit from actively choosing which comparative judgments to query?

Given that the proposed comparative learning framework allows non-linear models, the orthogonal query from Qian et al. [28] would not be applicable. We plan to adapt commonly used query strategies from active learning to better select which data pair to learn from next. With uncertainty sampling, the active learning framework for comparative learning is:

1. Randomly select K data pairs (i, j) and query the comparative oracles Y_{ij} .
2. Train the encoder $f_\theta(x)$ on all available comparative oracles Y_{ij} with hinge loss $\ell(Y_{ij}, C_{ij})$.
3. Apply $f_\theta(x)$ on all possible unlabeled data pairs and predict C_{ij} .
4. Select K data pairs with lowest $|C_{ij}|$. Query the comparative oracles Y_{ij} .
5. Repeat steps 2-4 until the predictions C_{ij} remain unchanged.

Evaluation Plan: Four treatments will be compared: the proposed comparative model with uncertainty sampling, the same model with random sampling, the LinearSVM-Comparative model from Qian et al. [28] with orthogonal query, and the regression model with uncertainty sampling.

Successful Criteria: RQ1.3 should be answered as yes when the human effort required by the comparative model with uncertainty sampling $t_c \times N_c$ is less than that with random sampling $t_c \times N_{rand}$, LinearSVM-Comparative [28] with orthogonal query $t_c \times N_l$, and that of the regression model with uncertainty sampling $t_r \times N_r$ when achieving similar prediction performance.

2.3 Thrust 2: Bias Reduction

The goal of Thrust 2 is to mitigate potential bias or discrimination against certain sensitive groups in predicting human decisions. As for the story point estimation case study shown in Figure 2, there are two types of backlog items in the Jira Software project: (1) internal user stories following the format “As a [ROLE], I would like ...” and (2) external defect fixing requests which do not follow the user story format. Underestimating the items of one type over the other one may cause severe and unfair consequences such as overloading certain developers or overlooking their work. Therefore, it is important to study how to effectively mitigate algorithmic bias in regression models such as the story point predictors.

2.3.1 Background

Algorithmic Bias Various different algorithmic bias notions [31, 32] have been defined to assess whether a trained machine learning model has ethical bias. When human decisions are considered unbiased, a perfect predictor of those decisions should always be unbiased as well. Many popular fairness testing metrics, for example FlipTest [33-35], individual fairness violation [36], and demographic parity [37], do not always allow a perfect predictor to be evaluated as fair and unbiased when the sensitive attributes are indeed correlated with the dependent variable [38]. This is why we specifically target separation [13] in the thrust. Separation requires the model’s prediction to be conditionally independent of the sensitive attribute given the ground truth labels – $C \perp A|Y$. In this way, separation always allows the perfect predictor $C = Y$ to be fair. In binary classification, separation requires the true positive rates and false positive rates to be the same across different sensitive groups – also known as equalized odds. Compared to the large number of bias mitigation algorithms being proposed to achieve equalized odds in classification [14, 39, 40], how to ensure separation in regression is underexplored. The most relevant work we find is that of Steinberg et al. [41] which directly optimizes for a mutual information-based metric during training. Such in-processing solutions require carefully chosen trade-off parameters and are not compatible with all regressors.

This is why we extend the Reweighting algorithm [14] and propose a novel preprocessing algorithm to ensure separation for any form of classifiers and regressors with both discrete and continuous sensitive attributes.

2.3.2 Proposed work

RQ2.1 How to ensure better separation for regression models?

Problem Statement. Consider training a model on a dataset $(X \in \mathbb{R}^n, A \in \mathbb{R}, Y \in \mathbb{R})$, the model will make predictions of $C \in \mathbb{R}$ based on its inputs (x, a) . The goal is to ensure the model prediction C satisfy the separation criterion $C \perp A|Y$.

Inspired by the Reweighting algorithm [14] for binary classification problems, we propose FairReweighting: **FairReweighting** adds a sample weight to rebalance each training data point

$$W(a, y) = \frac{P(a)P(y)}{P(a, y)}. \quad (2)$$

with $P(a)$, $P(y)$, and $P(a, y)$ estimated by their radius neighbor density:

$$\rho_N(x) = N(x_i \mid \text{dist}(x, x_i) < r) \quad \forall i \in \{1 \dots N\}. \quad (3)$$

Theorem 2.1. *Under the conditional independence assumption $X \perp A|Y$, a model trained with FairReweighting satisfies separation on its training data.*

Proof. To check whether separation is satisfied for the model on its training data, we have

$$\begin{aligned} P(C = c|y) &= \iint P(C = c|x, a)P(x, a|y)dxda \\ P(C = c|a, y) &= \int P(C = c|x, a)P(x|a, y)dx. \end{aligned} \quad (4)$$

This is because the model prediction C is completely decided by the input (x, a) and thus $C \perp Y|(X, A)$. Under conditional independence assumption of the data $X \perp A|Y$, we have

$$\begin{aligned} P(C = c|y) &= \iint P(C = c|x, a)P(x|y)P(a|y)dxda \\ P(C = c|a, y) &= \int P(C = c|x, a)P(x|y)dx. \end{aligned} \quad (5)$$

With FairReweighting in (2), the model learns from the weighted training data:

$$P(C = c|x, a) = \frac{P(Y = c, x, a)W(a, c)}{\int P(Y = y, x, a)W(a, y)dy}. \quad (6)$$

With (2) and the independence assumption of $X \perp A|Y$, we have

$$\begin{aligned} P(y, x, a)W(a, y) &= \frac{P(a)P(y)P(y, x, a)}{P(a, y)} \\ &= P(a)P(y)P(x|a, y) \\ &= P(a)P(y)P(x|y) = P(a)P(x, y). \end{aligned} \quad (7)$$

Apply (7) to (6) we have

$$\begin{aligned} P(C = c|x, a) &= \frac{P(a)P(x, c)}{\int P(a)P(x, y)dy} = \frac{P(x, c)}{\int P(x, y)dy} \\ &= \frac{P(x, c)}{P(x)} = P(c|x). \end{aligned} \quad (8)$$

Apply (8) to (5) we have

$$\begin{aligned}
P(C = c|a, y) &= \int P(c|x)P(x|y)dx \\
P(C = c|y) &= \iint P(c|x)P(x|y)P(a|y)dx da \\
&= \int P(c|x)P(x|y)dx \cdot \int P(a|y)da \\
&= \int P(c|x)P(x|y)dx = P(C = c|a, y).
\end{aligned} \tag{9}$$

Therefore, $C \perp A|Y$ and the model satisfies separation. $\square$

Evaluation Plan: Three treatments will be compared on regression datasets of story point estimation. These three treatments will include a baseline treatment without bias mitigation, the proposed FairReweighing algorithm, and a state-of-the-art in-processing algorithm [41]. We will use the mutual information-based metric proposed by Steinberg et al. [41] to measure the violation of separation of the regression models:

$$I[C; A | Y] = \int_Y \int_C \sum_{a \in \mathcal{A}} P(y, c, a) \log \frac{P(c, a | y)}{P(c | y)P(a | y)} dy dc \tag{10}$$

$$\hat{I}_{sep} = \frac{1}{n} \sum_{i=1}^n \log \frac{\rho(a_i | y_i, c_i)}{\rho(a_i | y_i)} \tag{11}$$

Successful Criteria: The proposed bias mitigation algorithm FairReweighing will be considered successful when it achieves better separation in terms of $\hat{I}_{sep}$ and maintains good prediction performance in terms of ρ and r_s compared to baselines. The preliminary results on the Jira Software project in Table 1 show that the proposed approach FairReweighing reduces the violation of separation while maintaining a good prediction performance.

Table 1: Preliminary results of RQ 2.1 on Jira Software.

Treatment	MAE	ρ	r_s	$\hat{I}_{sep}$
None	1.5504	0.4717	0.4335	0.0115
FairReweighing	1.5681	0.4580	0.4241	0.0074

RQ2.2 How to ensure better separation for comparative models?

Problem Statement. Consider training a model on a dataset of comparative judgments ($X_i, X_j \in \mathbb{R}^n, A_i, A_j \in \{0, 1\}, Y_{ij} \in \{1, -1\}$), the model will make predictions of $C \in \mathbb{R}$ based on its inputs (X, A) . The goal is to ensure the model prediction C satisfy the separation criterion $C \perp A|Y$ given a test data of direct ratings ($X \in \mathbb{R}^n, A \in \{0, 1\}, Y \in \mathbb{R}$).

With $A_{ij} = (A_i, A_j)$, we can generalize FairReweighing to comparative judgments:

$$W(a_{ij}, y_{ij}) = \frac{P(a_{ij})P(y_{ij})}{P(a_{ij}, y_{ij})}. \tag{12}$$

Since for comparative judgments $Y_{ij} = -Y_{ji}$, we have:

$$P(y_{ij}) = \frac{1}{2}, \quad \forall y_{ij} \in \{1, -1\}, \tag{13}$$

$$P(a_{ij}, y_{ij}) = \begin{cases} P(a_{ji}, -y_{ji}), & \forall y_{ij} \in \{1, -1\}, a_{ij} \in \{(1, 0), (0, 1)\} \\ \frac{1}{2}P(a_{ij}), & \forall y_{ij} \in \{1, -1\}, a_{ij} \in \{(0, 0), (1, 1)\} \end{cases}. \quad (14)$$

Therefore, we have

$$W(a_{ij}, y_{ij}) = \begin{cases} 1, & \forall y_{ij} \in \{1, -1\}, a_{ij} \in \{(0, 0), (1, 1)\} \\ \frac{P(a_{ij})}{2P(a_{ij}, y_{ij})}, & \forall y_{ij} \in \{1, -1\}, a_{ij} \in \{(1, 0), (0, 1)\} \end{cases}. \quad (15)$$

Evaluation Plan: To test whether the comparative FairReweighing algorithm in (15) improves separation for comparative models, we will test two treatments: 1) a baseline model trained on comparative judgments without bias mitigation and 2) the same comparative model trained on comparative judgments weighted by (15). Both models will be evaluated on a test set with known ground truth story points using the metric $\hat{I}_{sep}$.

Successful Criteria: The comparative FairReweighing algorithm proposed in (15) will be considered successful when it achieves a better separation in terms of $\hat{I}_{sep}$ and maintains a good prediction performance in terms of ρ and r_s compared to the baseline model.

2.4 Thrust 3: Bias evaluation in comparative judgments.

In addition to training the model on comparative judgments, this research thrust explores whether human effort can be further reduced by evaluating the model on a test set of comparative judgments. More specifically, this thrust analyzes how separation can be evaluated with comparative judgment oracles Y_{ij} instead of direct ratings Y .

2.4.1 Proposed work

Similarly to the separation criterion $C \perp A|Y$, we define comparative separation as follows:

Definition 2.2. *Given a test set of comparative judgments $(X_i, X_j \in \mathbb{R}^n, A_i, A_j \in \{0, 1\}, Y_{ij} \in \{1, -1\})$ and a prediction model $f_\theta(x)$, the model satisfies comparative separation if and only if*

$$C_{ij} \perp A_{ij}|Y_{ij} \quad (16)$$

where $C_{ij} = f_\theta(x_i) - f_\theta(x_j)$, $A_{ij} = (A_i, A_j)$.

RQ3.1 Is comparative separation equivalent to separation in binary classifications? As a special case, comparative separation is equivalent to separation in binary classification.

Theorem 2.3. *In binary classification problems where $Y, C \in \{0, 1\}$, $C_{ij} \perp A_{ij}|Y_{ij} \Leftrightarrow C \perp A|Y$.*

Proof. When $Y, C \in \{0, 1\}$, we have $C_{ij} \perp A_{ij}|Y_{ij}$ if and only if

$$\begin{aligned} P(C_{ij} = c_{ij}|A_{ij} = a_{ij}, Y_{ij} = y_{ij}) &= P(C_{ij} = c_{ij}|Y_{ij} = y_{ij}), \\ \forall c_{ij} \in \{-1, 0, 1\}, y_{ij} \in \{-1, 1\}, a_{ij} \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}. \end{aligned} \quad (17)$$

This is equivalent to

$$\begin{aligned} P(C_{ij} = 1|A_{ij} = a_{ij}, Y_{ij} = 1) &= P(C_{ij} = 1|Y_{ij} = 1) \\ \text{AND} \\ P(C_{ij} = -1|A_{ij} = a_{ij}, Y_{ij} = 1) &= P(C_{ij} = -1|Y_{ij} = 1), \\ \forall a_{ij} \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}, \end{aligned} \quad (18)$$

Table 2: Learning Objectives and Course Activities.

Learning Objective	Lecture Topics	Implementation Project Activities
O1 Why is human-AI collaboration important	Understand the cost of human decision-making, data collection and elicitation, IRB	Students will go through a mock IRB authoring & reviewing process and annotate 200 face images from SCUT-FBP5500 with their own preferences.
O2 How to prepare data for model training	Data cleaning, featurization, normalization, visualization, dimensionality reduction, and data balancing	Students will visualize their annotated data with histograms to show how they rate images with different demographics (race, gender). They will also prepare the image data for model training.
O3 How to efficiently train a model interactively with human feedback	Human feedback, active learning, and reinforcement learning	Students will train a VGG-16 model on 100 annotated images. The model is then tested on the rest 100 annotated images. Active learning will be implemented to sample a new batch of 10 images for the student to annotate. The model will then train on the newly annotated images and this process will be repeated for 5 times.
O4 How to evaluate and mitigate machine learning algorithmic bias	Individual fairness, group fairness, and mitigation of algorithmic bias from Thrust 2	The learned VGG-16 model will be tested for different group fairness metrics. Pre-, in-, post-processing techniques including FairReweighting will be implemented to mitigate the detected algorithmic biases.
O5 How to efficiently train a model with comparative judgments	Results from Thrust 1	Students will annotate 200 pairs of face images with their preferences. A comparative model with VGG-16 encoder is trained on 100 pairs and tested on the rest 100 pairs of the annotated images. Active learning is utilized to sample more pairs for annotation and update the comparative model for 5 rounds.
O6 How to evaluate and mitigate bias in comparative judgments	Results from Thrust 2 and 3	The model will be trained with comparative FairReweighting and will be tested for comparative separation.

aiming at improving any one of the activity results— e.g. design of a query strategy with better efficiency than uncertainty sampling or learn a model with higher accuracy than the VGG-16 model. This term project will allow students to explore what most interested them creatively. In return, the annotation data generated by the students in class will be utilized to further validate the results of the research findings in general decision-making processes. For example, the effort of direct annotation in **O1** and that of comparative judgments in **O5** from the same student will be recorded and compared to validate **RQ1.2** for whether comparative judgments save human effort. The performance of the model trained on direct ratings in **O3** will also be compared with that on comparative judgments in **O5** to validate **RQ1.1**. This will increase the impact of this work – around 30 students from the Master of AI program each year will be exposed to the cutting-edge research in this work.

3.2.2 Collaborative Software Engineering

Collaborative Software Engineering (SWEN-732) is a project-based learning (PjBL) course designed to teach students collaborative and modern software development practices. This course is offered once a year as a core course of the Master’s program in the Software Engineering Department at RIT. The course combines lectures with hands-on project work. Students work in teams throughout

the semester to design, implement, and deliver a complete software product from scratch, following an agile development process. Each group is responsible for managing requirements, planning iterations, performing sprint reviews, and continuously integrating and testing their code. During the planning phase, PI Yu will collaborate with Dr. Yiming Tang, the instructor of this course, to divide student teams into two groups: Group 1 will apply traditional strategies such as planning poker to estimate story point for each backlog item, while Group 2 will only provide comparative judgments on pairs of backlog items and apply comparative learning to rank the items. Surveys will be sent out to students to collect (1) the time required to provide direct estimates or comparative judgments, (2) whether the story points (either directly estimated or predicted by the comparative model) are useful in the downstream tasks, and (3) their overall feedback to the story point estimation pipeline. The survey result will be collected anonymously to analyze and answer RQ1.2.

3.2.3 NSF Research Experiences for Undergraduates

As a co-PI of an awarded NSF Research Experiences for Undergraduates program, PI Yu will actively involve undergraduate students in the research work proposed in this proposal (2 REU students are already involved in producing the preliminary results in 2025). This will broaden the impact of this work to 2-3 undergraduate students each year by engaging them in the research.

3.3 Course Evaluation

Cathleen Daniels Cerosaletti from the Warner School of Education at the University of Rochester will implement a structured data collection plan to ensure course objectives are met. She is an experienced STEM lead educational evaluator for multiple NIH (NIGMS supplement, T32, R25) and NSF programs (IUSE, FoW, SaTC, REU, IRES, ATE, NOYCE). This evaluation will focus on gathering feedback to assess educational outcomes [50] for the new content developed in the context of applied research findings and using distinct pedagogical methods. For educational interventions in a graduate-level course, the evaluator will implement curriculum review, surveys, and a focus group to collect mixed-methods data [51]. The evaluator will also assist with IRB documentation and approval for educational data collection.

Data Collection: To understand how research study results are implemented into the new course content, we ask: **EQ1** How does the course implementation leverage research findings?. The evaluator will undertake a curriculum review using a rubric based on the research plan to describe how research findings are implemented into the course design. To evaluate the amount and type of course impact on students, we ask: **EQ2** To what degree does the course implementation impact student learning? and **EQ3** In what ways does the course implementation impact student learning?. To respond to EQ2, data from post-lecture topic surveys (O1-O6) and a post-course survey will be collected as Likert-scaled ratings of perceived gains toward learning objectives, engagement, difficulty level of course activities, and feedback about course technology and pedagogical techniques. For each learning objective (O1-O6), an assessment will measure student achievement. Open-ended feedback will also be collected in all surveys to respond to EQ3. Post-course, a subset of graduate students will be invited to a focus group. The focus group will gather qualitative feedback including student narratives of the course experience to further address EQ3.

Data Analysis and Reporting: To characterize qualitative responses, themes will be developed from transcribed and aggregated data [52]. Likert-scaled quantitative data from surveys will be presented as descriptive statistics. All measures will be formative [53] in Year 4 to suggest further improvements in Year 5 and summative to report progress toward the objectives of the grant. In conjunction with grant timelines, reports will be provided to the PI after each data gathering effort.

4 Broader Impacts

Research. This work explores how to efficiently model SE decisions without introducing biases. With both training and test oracles being comparative judgments, modeling SE decisions would be more efficient and less biased. It will benefit various SE processes such as *story point estimation* and *vulnerability severity prediction*, as well as other cross-disciplinary processes such as: (1) *talent hiring and school admission* where HR persons/admission committees screen resumes to decide who get the interviews/admissions, (2) *crime sentencing* where judges decide the appropriate punishment based upon factors specific to the crime and the defendant, (3) *loan approval* where loan companies evaluate whether an applicant is qualified for a loan, and (4) *offensive language detection* where social media sites like Facebook or Twitter are monitored for hateful, offensive, or inaccurate language and removed when they violate use policies.

Education. As described in the education activities, the PI will incorporate the outcome from research to classrooms. The three education activities in Section 3.2 will advance the students’ knowledge in human-AI collaboration and AI4SE via project-based learning and impact around 50 students annually from the Master of SE and Master of AI programs at RIT as well as REU students from other universities.

Outreach. The PI has reached out to the Department of Software Engineering and the Department of Ph.D. in the Golisano College of Computing and Information Sciences at RIT to gain access to anonymized admission data. With human-AI collaboration results on these data, this work would potentially change the way departments at RIT admit students and hire new faculty and staff.

5 Project Timeline

The chart for completing the three research thrusts and education activities is shown below:

		Year 1	Year 2	Year 3	Year 4	Year 5
Thrust 1	Modeling comparative judgments					
	Human subject experiments with comparable judgments					
	Active learning with comparative judgments					
Thrust 2	Mitigating algorithmic bias in regression models					
	Mitigating algorithmic bias in comparative learning					
Thrust 3	Evaluate separation with comparative judgments					
Education	Collaborative Software Engineering					
	Research Method for AI					
	NSF REU Site					

6 Results from Prior NSF Support

PI Yu is the PI of an NSF-CRII award (#2245796), \$174,999; 06/01/2023 – 05/31/2026; titled “Testing Fairness in Human Decisions with Algorithmic Bias.” **Summary of Completed Work:**

Intellectual Merit: (1) Differential parity is designed to test relative bias between two sets of decisions; (2) FairBalance and FairReweight are developed to ensure separation in binary classification and regression models, respectively. These serve as the basis for this CAREER proposal.

Broader Impacts: The PI has developed and instructed a new course, *IDAI-720 Research Method for AI*, to integrate the results of this award. This course has exposed around 60 students (30 per year) to cutting-edge research in machine learning fairness. The work under this award also impacted an NSF REU site starting in 2025. **Publications Resulting from the Award:** So far, two publications [21, 54] in top SE journals have been produced under this award.

Research Products and Availability: All the source code and data generated under this award along with the course material of *Research Method for AI* are publicly available on GitHub. This award has also produced two preprints [55, 56] available on Arxiv.org submitted for publication.

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